

Project

HOTEL MANAGEMENT SYSTEM

Software Requirements Engineer (SW210)

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1 Introduction

1.1 Purpose: Mission Statement

To develop a secure, efficient, and a comprehensive hotel management software system by providing reservation and viewing services, payment, and improves customer service, ensures better management for reservation, and services like never before for having a grand vacation.

1.2 Scope of System

The “Hotel Management System,” herein referred to as “The System,” is software that helps manage hotel operations easily by automating simple and repeated tasks that take time but don’t need much human effort. It enables customers to book rooms, facilitates check-in and check-out processes while supporting both manual and online payments, and allows for registering and managing users. While the system won’t remove all human work, it ensures tasks are faster, more accurate, and improves the guest experience by maintaining data security, accuracy, and reliability.

2 User Characteristics

2.1 User/Stakeholder Profiles

Stakeholder	Interests	Constraints
Customer	Easy Booking and Payment.	None
Receptionists	Quickly booking and accurately management.	Old hardware that may affect system efficiency
Hotel Manager	Get detailed reports.	Reports must be current and easily printable.
Event Manager	Organize events.	Rooms may have limited availability.
Accountants	Keeping financial data safe. Prepare financial reports.	Data must be secure.
Data protection Authority	Ensuring the privacy and protection of customer data. Respecting privacy regulations.	Standards for data protection must be met by the system.
Maintenance team	Develop and maintain the system.	Limited time to complete the project.
IT Support Team	Fix technical issues quickly.	The system may be complex.
Payment Gateway Providers	Secure payment processing.	Must comply with payment standards.

3 Core System Requirements

3.1 Functional Requirements

3.1.1: The system shall allow the user to create an account using his email and setting an email and password.

3.1.2: The system shall allow the user to log in to his account and view the different kinds of rooms and sweets with its ratings and its companion services to make his reservation online.

3.1.3: The system shall allow the receptionist to handle the customers reservation by checking the requested dates and rates of the room.

3.1.4: The system shall ask the receptionist to enter the customer's personal details in order to accomplish the reservation.

3.1.5: The system shall allow the receptionist to cancel the reservation and update the status of room availability once it is canceled.

3.1.6: The system shall enable customers to send inquiries and specify the packages they are interested in or expect to use.

3.1.7: The system shall notify the users of the customers who made a reservation according to a query.

3.1.8: The system shall allow the external users to view the available room and available events' hall through the website.

3.1.9: The system shall give the user the choices for booking the events' hall the choices should include accommodations, meeting and banquets.

3.1.10: The system shall give the user the ability to edit his booking , by giving him the authority to change booking date and hall.

3.1.11: The system shall send a confirmation SMS message that includes the events' date and timing to the customer.

3.1.12: The system shall allow customers to specify the room type and the number of people for their booking.

3.1.13: The system shall allow customers to select a check-in date and then select a check-out date.

3.1.14: The system shall allow the user to choose the type of food that suits him such as Italian or Chinese and then display the menu according to his preferences.

3.1.15: The system shall allow staff to update the available menus daily, enabling customers to view the updated options and make reservations for food and beverages before their visit.

3.1.16: The system shall give the user the ability to customize the events hall, choose the preferred food and beverages, locate photoshoot locations.

3.1.17: The system shall allow online payments with its different types including Apple Pay, and PayPal.

3.1.18: The system shall allow the manager to print reports that documents all the details of the event including event time, date, decor , and menu.

3.1.19: The system shall link all payments to the reservation code issued for each booking, allowing customers to view the total bill and print the bill online.

3.1.20: The system shall maintain an up-to-date hotel website where visitors and customers can comment on photos of locations, food and beverages, and other facilities they have experienced.

3.1.21: The system shall be able to generate a payment report that includes all the details including payment date, amount paid, payment method, customer's name.

3.1.22: The system shall be able to generate a usage report that includes a list of available rooms.

4 Non-functional Requirements

4.1 Usability:

4.1.1 The system should enable users to achieve their goals effectively and efficiently.

4.1.2 Usability will be assessed based on three main attributes: Ease of Use, Learnability, and Flexibility.

4.2 Security:

4.2.1 The system should implement robust access control via MFA and safeguard sensitive information using the AES-256 encryption algorithm and Argon2 for hashing.

4.3 Accuracy:

4.3.1 The system should ensure that it provides precise and error-free performance in executing tasks and processes.

4.4 Reliability:

4.4.1 The system should ensure consistent performance, providing continuous protection without interruptions and maintaining reliability under various conditions.

4.4.2 The system's reliability should be assessed based on its ability to endure repetitive or high-intensity actions such as multiple, rapid successive attempts to initiate blocked calls-while maintaining consistent performance without any decline.

4.5 Maintainability:

- 4.5.1 The system should facilitate easy updates and maintenance to ensure long-term sustainability and functionality.
- 4.5.2 The system's maintainability will be assessed based on its ability to adapt to changes in technology and requirements, ease of debugging and issue resolution, and the level of effort required to perform routine maintenance tasks.